Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Data Pipelining, ETL Scheduling and Refresh, Complex SQL Queries, Table Joins, Aggregate Analytical Functions, Data Mining, Root Cause Analysis, Automated Task Simplification, Supply Chain & Operations Analytics, Stakeholder Engagement

Programming Languages & Tools: SQL, Python, R, AWS (S3, Redshift), Azure (Data Factory, Synapse, SQL), Snowflake, Apache Spark, PySpark, Power BI (DAX), Tableau, MS Excel, Git, Airflow

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Business Analyst & Data Operations Lead — **May 2022 – Present**

- **Architected complex high-performance SQL queries and ETL** jobs utilizing dimensional data modeling (Star/Snowflake schemas) and Slowly Changing Dimensions (SCD Type 1/2) to support ongoing reporting and analytics requirements.
- **Automated manual operational tasks and workflows** using Python scripts, PySpark, and agentic AI tools (KIRO IDE), cutting manual turnaround times by 50% and streamlining data extraction processes.
- **Designed and maintained scalable data pipelines and BI dashboards** in Power BI (DAX, measures) and Tableau to drive key business decisions and deliver executive KOM (Key Operating Metric) visibility.
- **Conducted deep-dive investigations and Root Cause Analysis** on complex reporting, BI, and data quality issues during high-demand periods, achieving 99.9% data availability SLA compliance.
- **Partnered closely with internal and external stakeholders** to translate business requirements into automated data quality audit frameworks and firewalls, reducing operational turnaround by over 80%.

Data Scientist & Analytics Consultant — **May 2022 – Present**

- **Developed proactive anomaly detection and predictive models** using XGBoost, Random Forests, and Scikit-Learn to improve system reliability by 20% and mitigate operational discrepancies.
- **Implemented advanced SARIMA forecasting models** for customer behavior and resource trends, optimizing capacity planning and cutting response times by 20%.
- **Built automated quality testing frameworks and firewalls** using Python and SQL to ensure data accuracy for newly ingested operational data and resources.

Information Systems & Data Science Researcher — **May 2022 – Present**

- **Conducted advanced statistical analysis and data mining** utilizing relational databases and scripting languages to extract actionable business insights from complex datasets.
- **Designed relational database schemas and optimized data extraction** procedures to support large-scale enterprise reporting initiatives.
- **Streamlined data pipeline performance and query execution** through rigorous performance tuning and efficient indexing strategies.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.

- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.